

Linear Algebra-Midterm

Student Name/Number: _____

1. Solve the following system of linear equations by row reducing the associated augmented matrix. Find the row reduced echelon form of the matrix. Write the solution set and find two particular solutions for the system if there exist more than one solution.

$$\begin{aligned}x_1 + 2x_3 - 2x_5 &= 11 \\x_2 - 3x_4 - x_5 &= 0 \\2x_1 + 5x_3 - 4x_5 &= 27 \\3x_1 + 2x_2 + 7x_3 - 6x_4 - 8x_5 &= 38\end{aligned}$$

2. Find values of c such that the system of linear equations has (a) exactly one solution, (b) an infinite number of solutions, and (c) no solution.

$$\begin{aligned}x_1 + x_2 + cx_3 + x_4 &= c \\-x_2 + x_3 + 2x_4 &= 0 \\x_1 + 2x_2 + x_3 - x_4 &= -c\end{aligned}$$

3. For which values of a the following system of linear equations will have no solutions, exactly one solution, and infinitely many solutions?

$$\begin{aligned}x + 2y - 3z &= 4, \\3x - y + 5z &= 2, \\4x + y + (a^2 - 14)z &= a + 2,\end{aligned}$$

4. Solve the following linear system

$$\begin{aligned}2x + y &= 1, \\y - z &= 2, \\-2x + y + z &= -2,\end{aligned}$$

by finding the LU-factorization of the coefficient matrix.

5. Find inverses, if they exist, for the following matrices:

$$A = \begin{bmatrix} 2 & 0 & -1 & 3 \\ 1 & -2 & 3 & 1 \\ 4 & 1 & 0 & -1 \\ 1 & 3 & -2 & -5 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & -1 & 5 \\ -1 & 1 & 0 & -3 \\ 0 & 2 & -3 & 7 \\ 2 & 1 & -2 & 12 \end{bmatrix}$$

6. Find the inverses of the following matrices by using elementary matrices.

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 6 & -1 \\ 0 & 0 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

7. Use elementary matrices to find the inverse of

$$A = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & c \end{bmatrix}, c \neq 0$$

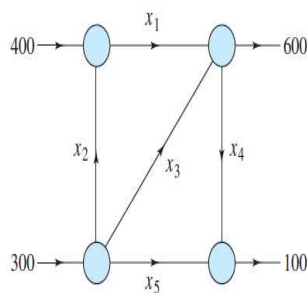
8. Determine the polynomial $p(x) = a_0 + a_1x + a_2x^2$ whose graph passes through the points $(1, 4)$, $(2, 0)$, $(3, 12)$.

9. (25pt) Networks composed of branches and junctions are used as models in many diverse fields such as economics, traffic analysis, and electrical engineering. In such models it is assumed that the total flow into a junction is equal to the total flow out of the junction.

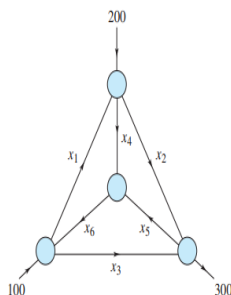
The flow of traffic (in vehicles per hour) through a network of streets is shown in the Figure.

(a) Solve this system for x_i , $i = 1, 2, 3, 4, 5, 6$.

(b) Find the traffic flow when $x_3 = 1000$, $x_5 = 50$ and $x_6 = 50$.



10. (i) Solve this system for x_i , $i = 1, 2, 3, 4, 5$. (ii) Find the traffic flow when $x_3 = 0$ and $x_5 = 1000$.



11. Determine the currents I_1, I_2, I_3, I_4, I_5 and I_6 for the electrical network shown in the following Figure.

